

Summary Report - SIH2026ppt.pdf

AEGIS Sovereign Intelligence Report | Generated: 2026-09-02 14:58:54 UTC | Classification: **CONFIDENTIAL**
INTERNAL

EXECUTIVE SUMMARY

Aegis proposes the 'Sovereign AI Workbench,' a 100% on-premise AI solution designed for confidential industrial work. The system utilizes open-weight, multimodal LLMs, offering intelligent model routing, agentic workflows (Plan → Execute → Verify → Re-plan), multimodal document understanding (PDFs, images, scans, drawings), and local RAG over organizational knowledge. It emphasizes data sovereignty, secure execution within a sandbox environment, and delivers verified outputs (reports, code) while dynamically loading models based on task requirements. The solution aims to automate repetitive tasks and improve productivity within industrial workflows, drawing inspiration from existing sovereign AI platforms like Siemens, Palantir, and IBM Watsonx.

KEY FINDINGS

The Sovereign AI Workbench prioritizes data residency and security by executing entirely on-premise, eliminating cloud API calls. It employs an agentic architecture enabling autonomous task completion through a defined workflow (Plan, Execute, Verify, Re-plan). Multimodal input support is core, processing text, PDFs, scans, images, and drawings. The system features dynamic model loading—'Dynamic Model Loading'—to optimize resource utilization by loading models on-demand. A key differentiator is the modular model architecture, facilitating easy addition or swapping of AI models. The team leverages quantized LLMs (7B/8B Class) for efficiency.

DETAILED ANALYSIS

The proposed system utilizes a secure web UI for user interaction, integrating upload and query functionality. The Agent Controller drives task understanding and planning, utilizing a task-aware selection mechanism. Document workflows leverage RAG and validation, while coding workflows enable safe code generation and execution within a sandbox. A model registry manages dynamic model loading, selecting appropriate models based on the task at hand. The system's architecture addresses potential challenges like GPU limitations and data leakage through strategies such as network isolation, audit logs, and source grounding. The team acknowledges the importance of hallucination mitigation through rigorous validation processes and safeguards.

RISKS AND OPERATIONAL ISSUES

Potential risks include GPU limits and contention leading to performance bottlenecks. The OCR + VLM + Document pipeline introduces complexity requiring careful monitoring and validation. Data leakage remains a concern necessitating robust security measures like network isolation and audit trails. The reliance on quantized models requires ongoing evaluation to maintain accuracy. The system's air-gapped nature presents challenges related to software updates and potential maintenance needs. The team recognizes the need for ongoing model governance and selection.

RECOMMENDATIONS

To mitigate risks, the team recommends a phased deployment, beginning with controlled environments, alongside continuous monitoring of GPU utilization and model performance. Investing in robust validation mechanisms is crucial for addressing hallucination. Implementation of comprehensive audit logging and data loss prevention measures is strongly advised. Exploring integration with existing data governance frameworks is recommended to ensure compliance and maintain a secure environment.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
SIH2026ppt.pdf	Page(s) 1, 2, 3, 4, 5, 6, 7	High

Verification Integrity Hash: ea0e123cdaad29285491a18689dc7703 . . .